

SAPPHIRE

INVENTORY TRACKER

Module Design & User Guide

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1. Module Overview

The **Inventory Tracker** is a mobile-based utility developed under the **My Sapphire portal**, specifically designed for **store employees** to instantly scan barcodes of articles directly on the sales floor. This eliminates unnecessary traffic and dependency on counter terminals for inventory lookups, improving store efficiency and customer service.

2. Objective

To provide store staff a quick, accurate, and user-friendly solution to check stock availability of articles across multiple store locations through barcode scanning.

3. Design & Functionalities

3.1 Platform Scope

Feature	Mobile App	Web Portal
Scan Barcode	✓	✗
View Inventory Details	✓	✓
Assign Store Access	✓	✓

***Note:** Barcode scanning is **exclusive** to the **mobile app**. The web portal supports admin-level functions such as user role assignment and reporting (if enabled).*

3.2 Barcode and SKU System

- When a **barcode** is scanned, the system fetches the associated **SKU (Stock Keeping Unit)** — a **system-generated unique identifier** for that article.
- This ensures that each article is uniquely identifiable and the correct inventory information is retrieved.

3.3 Key Functions

When a store employee **scans the barcode** using the mobile app, the system displays the following fields:

Field	Description
Current (Store)	The store where the scan is being made
Location	Back Store Location
Price	Selling price of the scanned article
Barcode	Unique identification code printed on article tags
Matching Separate	Article wise quantity of matching separates
Size Sets	Size wise quantity of matching separates
Size	Available sizes with quantity breakdown
Discount (Disc.)	Current active discount (if any)
Online Store	Online availability of the respective articles
Other Locations	Size wise available quantity of same SKU and its MS

3.4 Role & Access Management

- A dedicated role named "**Inventory Tracker**" is created in the system.
 - This role is **assigned to store employees** through the admin interface or assigning application.
-

3.5 Location-Based Availability Logic

- When an article is scanned, the app also displays **other outlets in the same city** where the article is available.
- The results are shown based on:
 - **City match** of the current location
 - **Nearest outlet proximity** (using store address or geo-coordinates)

3.6 Current Store Visibility

When store staff scan the barcode, their respective stores are shown on priority basis.

3.7 Error Handling Scenarios

Scenario	Message
Invalid barcode	"Article not found. Please scan again."
No inventory available	"Out of stock at this location."
User not assigned to any store	"No store access found. Contact Admin."
Network issues	"Connection failed. Please try again later."

3.8 Security and Restrictions

- Secure login with user authentication.
-

4. Module Interface

Step-by-Step Journey with Screenshots

A. Login Screen

- The user enters their credentials.
- Only users with the assigned **Inventory Tracker role** can access the scanning feature.

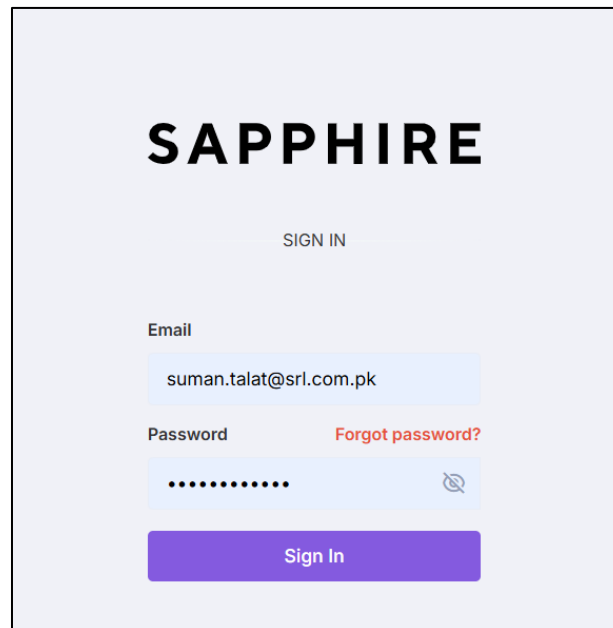
A screenshot of the SAPHIRE login interface. The word "SAPHIRE" is displayed in large, bold, black capital letters at the top. Below it, the text "SIGN IN" is centered in a smaller, black, sans-serif font. There are two input fields: the first is labeled "Email" and contains the text "suman.talat@srl.com.pk"; the second is labeled "Password" and contains a series of dots, with a "Forgot password?" link in red text to its right. A purple "Sign In" button is located at the bottom of the form.

Figure 1 Login Credentials

B. Inventory Tracker – Web View

The web version of the **Inventory Tracker** in the **My Sapphire** portal allows store employees to **track inventory by manually entering barcodes**.

Although barcode scanning is only available on the mobile app.

When navigating to the **Inventory Tracker** module under the **My Sapphire** portal, two functional forms are available:

1. **Scan Barcode** – Allows users to manually enter a barcode to check inventory.
2. **Product List** – Displays a list of available articles along with their inventory details.

These forms are accessible under the **Inventory Tracker functionality** and are visible based on the user's assigned role and permissions.

IT/Admin teams use this screen to manage user access and interface visibility based on user requirements.

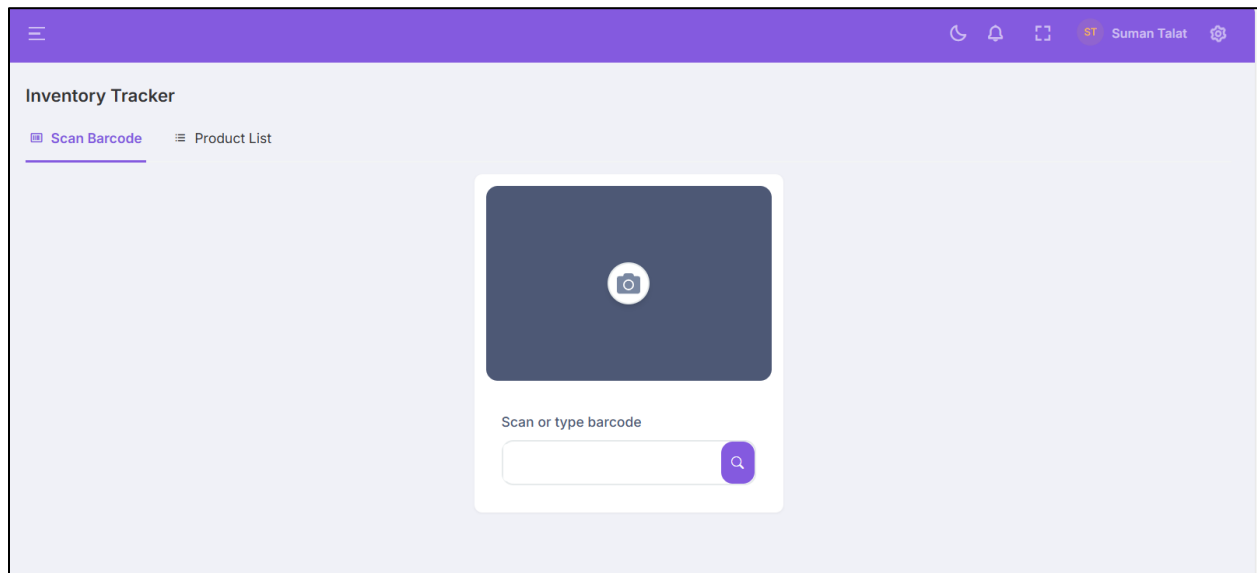


Figure 2. Inventory Tracker – Web Interface displaying manual barcode input. Actual scanning is supported only on mobile devices.

C. Enter Barcode to view Inventory

Once the user selects the **Barcode** form under the **Inventory Tracker module**, they can proceed to **manually enter the barcode** of the desired article.

- The barcode can be typed directly into the input field.

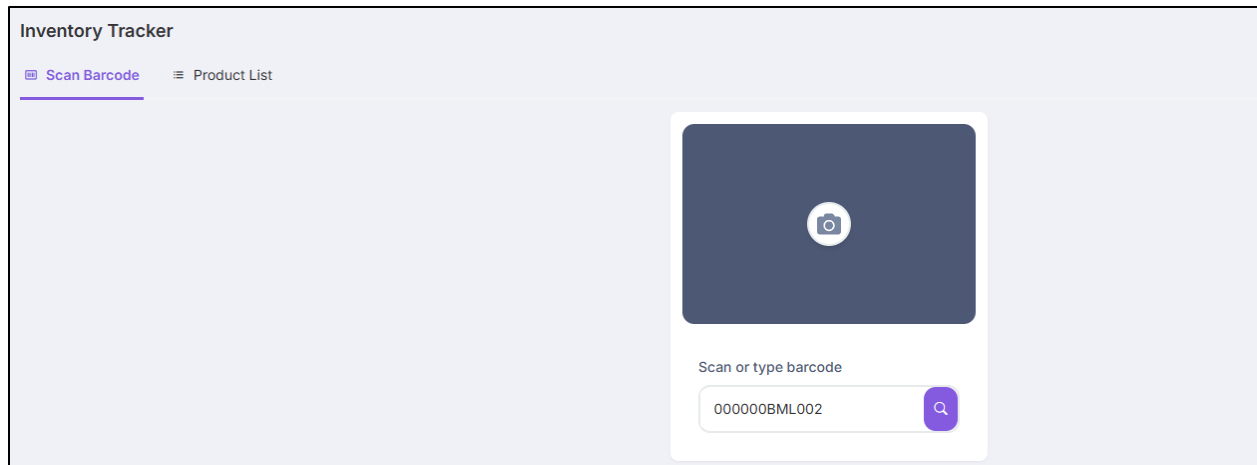


Figure 3. Inventory Tracker – Entering barcode “000000BML002”

Once the barcode is entered, the system retrieves and displays the inventory details of the respective article.

Displayed Information Includes:

- **Current Store:** The store where the query is being made (e.g., *Amana Mall, Lahore*)
- **Available Quantity:** Shown next to the store name (e.g., *Qty: 41*)
- **Price:** Retail selling price of the item
- **Barcode:** Entered barcode of the article
- **MS:** If part of any bundle offer
- **Size Set:** Size wise quantity of matching separates
- **Size & Discount:** Available size details and any applied discount
- **Other Locations:** Option to view inventory of the same article in other outlets across the same city
- **Online Location:** Online availability of the respective articles.

Buttons for **Re-scan** and **Clear Search** are also available to quickly reset the input.

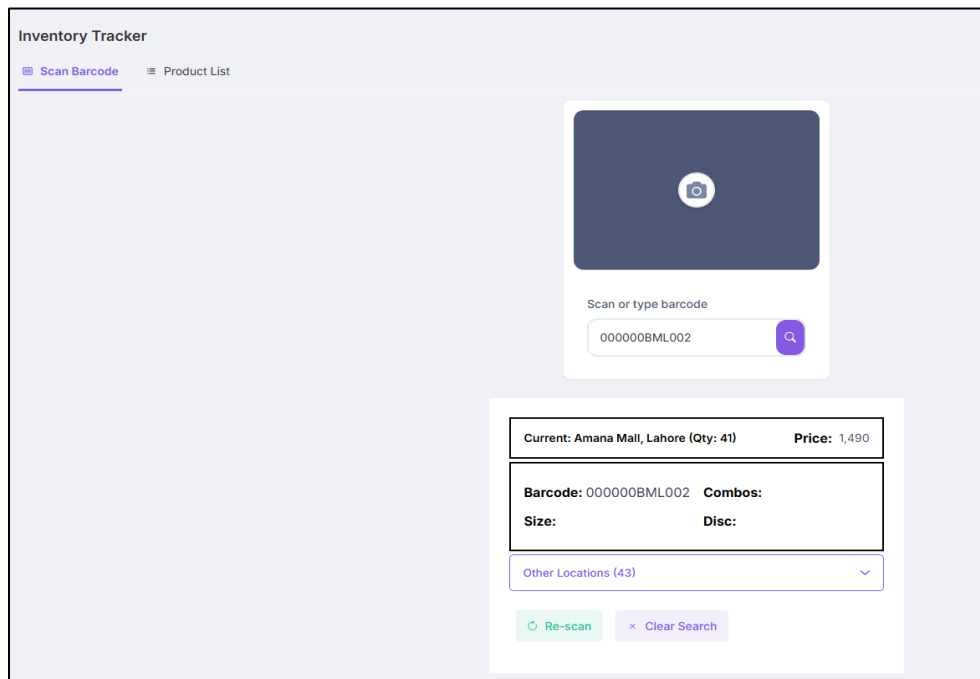


Figure 4. Inventory Tracker – Inventory result displayed after entering barcode “000000BML002”

By clicking on the “**Other Locations**” section. The system prioritizes and displays **outlets within the same city**.

Displayed Attributes Include:

- **Store Name** – Outlet name where stock is available
- **Quantity (Qty)** – Total number of units available at that outlet
- **Sizes** – Size-wise availability
- **MS (Matching Separate)** – Displays the stock of matching articles related to the scanned item
- **Order on Behalf** – Indicates whether the store allows ordering the article for the customer from another outlet (if enabled)

Store	Quantity	Sizes	Combos	Order On Behalf
 UOL FOL, Lahore	12	N/A	N/A	
 Bahadurabad Store	12	N/A	N/A	
 Square One Mall, Karachi	13	N/A	N/A	
 Y-Block DHA, Lahore	12	N/A	N/A	
 Fortress Square Mall, Lahore	12	N/A	N/A	
 Allama Iqbal Town, Lahore	16	N/A	N/A	
 Dolmen Mall Lahore	15	N/A	N/A	
 Emporium Mall, Lahore	34	N/A	N/A	
 Packages Mall, Lahore	16	N/A	N/A	
 PWD Islamabad Store	12	N/A	N/A	

Figure 5. Other Locations with details of stock, sizes, MS and order-on-behalf option.

D. Product List Form

The **Product List** is the second available form under the **Inventory Tracker** module. Unlike the barcode scan form, which displays information for a specific article, the Product List provides a **complete view of all inventory items** accessible to the user based on their assigned store(s).

Displayed Columns Include:

- **Barcode** – Unique code assigned to each article
- **Warehouse Name** – Name of the store or warehouse where the stock is located
- **Warehouse Code** – System-generated identifier for the warehouse
- **Sales Price** – Standard selling price of the article
- **On-Hand Quantity** – Total quantity currently available in stock
- **Discounted Price** – Final price after applying active discount
- **Discount Percentage** – Percentage of discount applied (if any)
- **Size Set** – Indicates size grouping or configuration (e.g., S-XL, Free Size)
- **Size** – Individual size of the article (e.g., M, L, XL)
- **MS (Matching Separate)** – Breakdown of matching pieces (Shirt, Dupatta, Trouser), e.g., S3 D1 T2

This form helps store staff and supervisors **view, filter, and verify inventory** details across assigned stores without the need for scanning, and can be used as a quick reference during daily operations or stock checks.

ST Suman Talat

Inventory Tracker

Scan Barcode

Product List

Show 10

Columns

Search Here

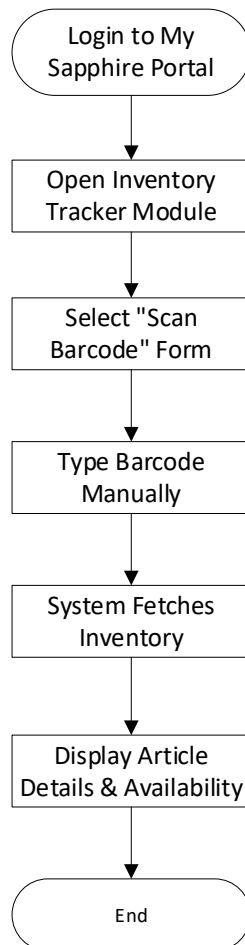
Barcode	Ware House Name	Ware House Code	Sales Price	Onhand Qty	Discount Price	Discount Per	Size Set	Size	Combo
DEPRW25V155D	Safa Gold Mall, Islamabad	S3011SFA	NaN	N/A	N/A	N/A	N/A	N/A	D0
002SDF24V41D	Square One Mall, Karachi	S5010SOM	NaN	N/A	N/A	N/A	N/A	N/A	D0
DEPRW25V155D	Kutchery road, Gujrat	S4004GRT	NaN	N/A	N/A	N/A	N/A	N/A	D0
DEPRW25V155D	Allama Iqbal Town, Lahore	S2001AIT	NaN	N/A	N/A	N/A	N/A	N/A	D0
DEPRW25V155D	Gulberg II, Lahore	S2005GUL	NaN	N/A	N/A	N/A	N/A	N/A	D0
3PRDYE25V17D	Square One Mall, Karachi	S5010SOM	NaN	N/A	N/A	N/A	N/A	N/A	D0
3PBEW24V620D	Gulshan KDA, Karachi	S5006KDA	NaN	N/A	N/A	N/A	N/A	N/A	D0
3SSFJQDW248D	Square One Mall, Karachi	S5010SOM	NaN	N/A	N/A	N/A	N/A	N/A	D0

Figure 6. Product List form displaying inventory

5. Flowcharts

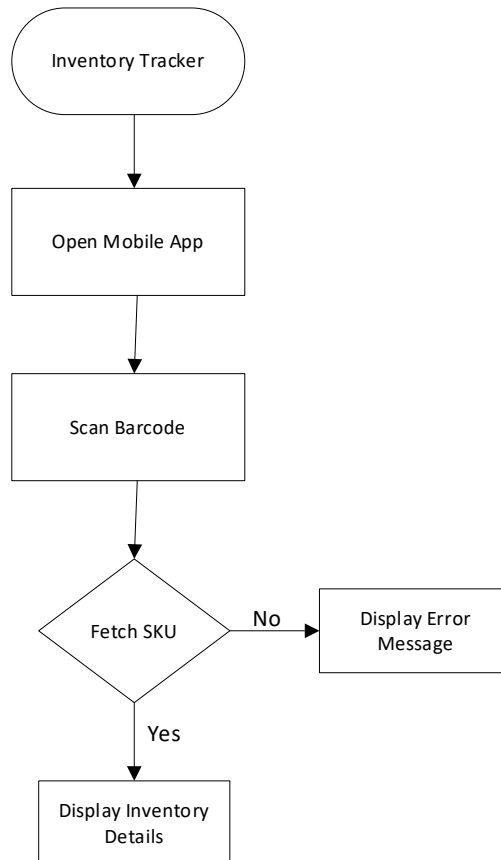
5.1 Web Flow

This version allows users to **manually enter barcodes** to check inventory and access the full **product list**. Scanning is not supported on the web.



5.2 Flowchart for Mobile Application:

The following flowchart illustrates the step-by-step process for store employees using the Inventory Tracker mobile application, from secure login to barcode scanning and viewing inventory details at other locations. It is designed to provide a clear overview of the expected user journey and ensure consistent usage across all stores.



For further assistance or enhancements, please contact the IT/Admin team.